

Signorini's Problem

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Abstract

This is the abstract.[1]

1 Introduction

We are given a body $\Omega \subset \mathbb{R}^d$. While all of the below may be formulated for arbitrary $d \geq 1$, it is of course reasonable to restrict our selfs to the real world application $d \in \{2, 3\}$. Further necessary topological assumptions on Ω are introduced in the next subsection. This body may be sitting on a *rigid foundation*. See Figure 1 (left). Furthermore, let our body be made of an *elastic material*, allowing Ω to deform due to *internal* and external *forces*. If these forces are large enough, Ω in its initial state as in Figure 1 (left) is not in a *balanced* configuration. The internal and external forces are not in equilibrium. If we now allow all forces to act until the body is in an equilibrium configuration, this could look like Figure 1 (right). The body Omega in this state is designated $\varphi(\Omega)$.

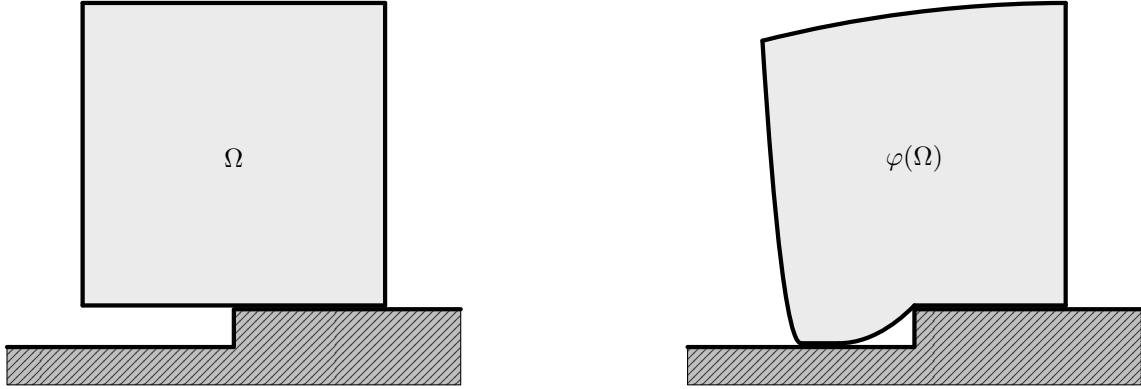


Figure 1: Body Ω on a rigid foundation in initial (left) and equilibrium (right) configuration.

The transition of Ω in its initial configuration into the Ω in its equilibrium configuration under prescribed internal and external forces defines the map

$$\varphi : \Omega \rightarrow \mathbb{R}^d, \quad \varphi(x) = x + u(x),$$

where $u : \Omega \rightarrow \mathbb{R}^d$ is called the *displacement* function. The Signorini Problem seeks precisely this function under given elasticity properties, forces, boundary conditions and other rules.

2 Geometrical and Physical Setup

Naturally, all of the above must be mathematically modeled in order to apply suitable theories and methods, e.g. from functional analysis. This will be covered in the following sections. Throughout, $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ denotes an open, bounded and connected domain with boundary $\Gamma := \partial\Omega$, that is at least Lipschitz, i.e. $\Gamma \in \mathcal{C}^{m,1}$, $m \geq 0$. Among other things, this provides us with an *outward normal* $\mathbf{n} \in \mathcal{C}^{m-1,1}(\Gamma; \mathbb{R}^d)$ if $m = 1$. For now, $m = 0$ is sufficient. Then, $\mathbf{n} \in L^\infty(\Omega; \mathbb{R}^d)$. Later on, higher regularity on Γ is needed. We begin by dividing the boundary into different sub-boundaries, which will later each be subject to different boundary conditions. Then the elasticity properties of the body are appropriately modeled and we discuss how the acting forces are understood. Finally, it is defined which shift functions are admissible at all.

2.1 Boundary Decomposition

The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}, \quad |\Gamma_C|, |\Gamma_D| > 0, \quad |\Gamma_N| \geq 0.$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 2).

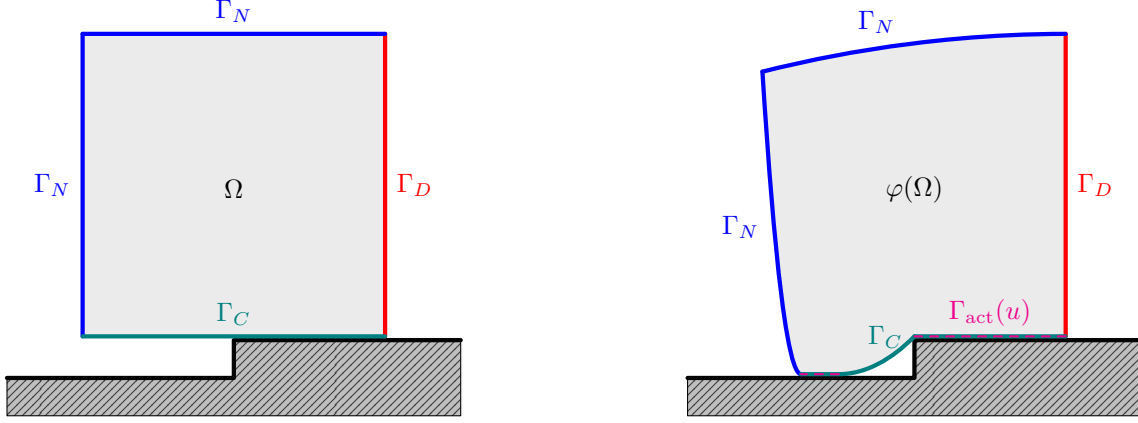


Figure 2: Decomposition of Γ in Dirichlet, Neumann and Contact parts before (left) and after displacement (right).

While Γ_C and Γ_D are assumed to be of positive $(d - 1)$ -Lebesgue measure, Γ_N may in some instances be a null set, or more roughly speaking, nonexistent. On Γ_D , we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D . The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C . While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\text{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u . Later on, the boundary conditions imposed on Γ_C enforce *unilateral contact* through *nonpenetration* and *complementarity*. The rigid foundation can be modeled through a *gap-function* $g : \Gamma_C \rightarrow \mathbb{R}$, describing the distance between Γ_C and the foundation. While the foundation in Figure 2 suggests a piece-wise constant gap-function g , we will throughout the whole report assume $g = 0$.

2.2 Basic Notations on Sobolev Spaces

This subsection is intended to fix some of the notations used in this report. Since Signorini's problem is formulated as a search for the displacement function as a partial differential equation, Sobolev spaces are of particular importance. The Sobolev space $W^{k,p}(\Omega; \mathbb{R}^d)$ contains every $L^p(\Omega; \mathbb{R}^d)$ -function, such that weak derivatives up to order $k \geq 0$ are again in $L^p(\Omega; \mathbb{R}^d)$. Here, we are only interested in the Hilbert space case $p = 2$ and we write $H^k(\Omega; \mathbb{R}^d) := W^{k,2}(\Omega; \mathbb{R}^d)$. The L^2 -inner product and norm is denoted by $(\cdot, \cdot)_{0,\Omega}$ and $\|\cdot\|_{0,\Omega}$. We record this introduction in the following definition and simultaneously establish a notation for the associated norm.

Definition 2.1 (Sobolev Spaces of Integer Order). *Let $k \in \mathbb{N}_0$. The Sobolev space $H^k(\Omega; \mathbb{R}^d)$ is defined by*

$$H^k(\Omega; \mathbb{R}^d) := \left\{ u \in L^2(\Omega; \mathbb{R}^d) \mid D^\alpha u \in L^2(\Omega; \mathbb{R}^d) \ \forall \alpha \in \mathbb{N}_0^d, \ |\alpha| \leq k \right\},$$

where $D^\alpha u$ is understood in the weak sense. Moreover, we equip $H^k(\Omega; \mathbb{R}^d)$ with the norm

$$\|u\|_{k,\Omega} := \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{0,\Omega}^2 \right)^{1/2}, \quad u \in H^k(\Omega; \mathbb{R}^d).$$

We adopt the usual convention $H^0(\Omega; \mathbb{R}^d) = L^2(\Omega; \mathbb{R}^d)$ explaining the notion of the L^2 -norm. Furthermore, we state that $H^k(\Omega; \mathbb{R}^d)$ is a Hilbert space.

Proposition 2.1 ($H^k(\Omega; \mathbb{R}^d)$ is a Hilbert Space). *The Sobolev space $H^k(\Omega; \mathbb{R}^d)$ together with the inner product*

$$(u, v)_{k,\Omega} := \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{0,\Omega}, \quad u, v \in H^k(\Omega; \mathbb{R}^d)$$

is a Hilbert space and the induced the norm $\|\cdot\|_{k,\Omega}$. □

After the integer-order spaces $H^k(\Omega; \mathbb{R}^d)$, we also need Sobolev spaces of non-integer order. The main reason is that trace operators naturally lead to boundary spaces of fractional order. In particular, traces of $H^1(\Omega; \mathbb{R}^d)$ -functions are, in general, not measured in another integer-order Sobolev space, but in a space of order $1/2$ on the boundary and similarly on boundary parts $\Gamma_0 \subset \Gamma$. For a non-integer order $s = k + \theta$ with $\theta \in (0, 1)$, the remaining fractional part θ is measured by difference quotients. More precisely, one measures how strongly a function oscillates between pairs of points $x, y \in \Omega$, with a weight that becomes singular as $|x - y| \rightarrow 0$. This leads to the following quantity.

Definition 2.2 (Slobodeckij Seminorm). *Let $m \geq 1$ and $\theta \in (0, 1)$. For a measurable function $u : \Omega \rightarrow \mathbb{R}^d$, we define the Slobodeckij semi-norm*

$$|u|_{\theta,\Omega} := \left(\int_{\Omega} \int_{\Omega} \frac{|u(x) - u(y)|^2}{|x - y|^{d+2\theta}} dx dy \right)^{1/2},$$

whenever the right-hand side is finite.

The quantity $|u|_{\theta,\Omega}$ is only a semi-norm, since it vanishes on constant functions. Combined with an L^2 -term, however, it yields a genuine norm. With this, we can define Sobolev spaces of fractional order and just as above, state, that this provides us with a Hilbert space together with suitable inner product.

Definition 2.3 (Sobolev Spaces of Fractional Order). *Let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. The Sobolev space of fractional order s is defined by*

$$H^s(\Omega; \mathbb{R}^d) := \left\{ u \in H^k(\Omega; \mathbb{R}^d) \mid |D^\alpha u|_{\theta,\Omega} < \infty \ \forall \alpha \in \mathbb{N}_0^d, \ |\alpha| = k \right\},$$

where $D^\alpha u$ is understood in the weak sense. Moreover, we equip $H^s(\Omega; \mathbb{R}^d)$ with the norm

$$\|u\|_{s,\Omega} := \left(\|u\|_{k,\Omega}^2 + |u|_{s,\Omega}^2 \right)^{1/2}, \quad u \in H^s(\Omega; \mathbb{R}^d).$$

Proposition 2.2 ($H^s(\Omega; \mathbb{R}^d)$ is a Hilbert Space). *Let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. Then $H^s(\Omega; \mathbb{R}^d)$ is a Hilbert space with inner product*

$$(u, v)_{s,\Omega} := (u, v)_{k,\Omega} + \sum_{|\alpha|=k} \int_{\Omega} \int_{\Omega} \frac{(D^\alpha u(x) - D^\alpha u(y)) \cdot (D^\alpha v(x) - D^\alpha v(y))}{|x - y|^{d+2\theta}} dx dy$$

for $u, v \in H^s(\Omega; \mathbb{R}^d)$. The induced norm is $\|\cdot\|_{s,\Omega}$. □

For $u \in H^s(\Omega; \mathbb{R}^d)$, pointwise boundary values are in general not defined. The trace theorem provides a canonical replacement.

Definition 2.4 (Classical Trace on Smooth Functions). *We define the classical trace operator*

$$\gamma_0 : C^\infty(\overline{\Omega}; \mathbb{R}^d) \rightarrow L^2(\Gamma; \mathbb{R}^d), \quad \gamma_0 u := u|_\Gamma$$

on the space

$$C^\infty(\overline{\Omega}; \mathbb{R}^d) := \left\{ u|_{\overline{\Omega}} \mid u \in C^\infty(U; \mathbb{R}^d) \text{ for some open neighborhood } U \supset \overline{\Omega} \right\}.$$

Definition 2.5 (Boundary Sobolev Spaces). *Let $s > 1/2$. We define*

$$H^{s-1/2}(\Gamma; \mathbb{R}^d) := \gamma_0(C^\infty(\overline{\Omega}; \mathbb{R}^d))$$

equipped with the norm

$$\|g\|_{s-1/2, \Gamma} := \inf \left\{ \|u\|_{s, \Omega} \mid u \in C^\infty(\overline{\Omega}; \mathbb{R}^d), \gamma_0 u = g \right\}.$$

Notice how the underlying set is the same for different values of $s > 1/2$, but it is equipped with different norms.

Theorem 2.1 (Trace Theorem). *Let $\Gamma \in \mathcal{C}^{m,1}$, $m \geq 0$ and $1/2 < s \leq m+1$. Then the classical trace operator γ_0 extends uniquely to a continuous surjective linear operator*

$$\gamma : H^s(\Omega; \mathbb{R}^d) \rightarrow H^{s-1/2}(\Gamma; \mathbb{R}^d).$$

In particular, there exists a constant $C_{\text{tr}} > 0$ such that

$$\|\gamma u\|_{s-1/2, \Gamma} \leq C_{\text{tr}} \|u\|_{s, \Omega} \quad \forall u \in H^s(\Omega; \mathbb{R}^d).$$

□

In the special case $s = 1$, we further have

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; \mathbb{R}^d)}^{\|\cdot\|_{1, \Omega}} =: H_0^1(\Omega; \mathbb{R}^d),$$

where $\mathcal{D}(\Omega; \mathbb{R}^d)$ denotes the space of $\mathbb{R}^d$ -valued test functions on Ω .

2.3 Elasticity and Forces

In Elasticity, stresses are described by *symmetric second order tensors*,

$$\mathbb{S}^d := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \mid s^\top = s \right\}.$$

We identify the tensor space by symmetric $d \times d$ -matrices in the canonical basis. Indeed,

$$\mathbb{S}^d \cong \left\{ S \in \mathbb{R}^{d \times d} \mid S^\top = S \right\}.$$

As an inner product on $\mathbb{R}^{d \times d}$, we use the standard Frobenius product

$$A : B := \sum_{j=1}^d \sum_{k=1}^d A_{jk} B_{jk}, \quad |A| := \sqrt{A : A}$$

for real $d \times d$ -matrices A and B , while for $x, y \in \mathbb{R}^d$ we denote by

$$x \cdot y := \sum_{j=1}^d x_j y_j, \quad |x| := \sqrt{x \cdot x},$$

the Euclidean inner product and norm. As $\mathbb{S}^d$ may be identified with a subspace of $\mathbb{R}^{d^2}$, the above definitions of Sobolev spaces may be naturally extended to $\mathbb{S}^d$ -valued functions on Ω . We continue by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega; \mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\bar{\Omega}$ by u is quantified.

Definition 2.6 (Small Strain Tensor). *The (small) strain tensor ε is defined by*

$$\varepsilon : H^s(\Omega; \mathbb{R}^d) \rightarrow H^{s-1}(\Omega; \mathbb{S}^d), \quad u \mapsto \varepsilon(u) := \frac{1}{2} (\nabla u + \nabla u^\top).$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* $\mathcal{A}$, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field $\mathcal{A}$ is prescribed by the material and may vary in space, describing a *heterogeneous* body. For a fourth order tensor $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{kl}, \quad s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

Definition 2.7 (Elasticity Tensor Field). *An elasticity tensor field is a fourth order tensor*

$$\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$$

such that the coefficients satisfy the symmetries

$$\mathcal{A}_{ijkl}(x) = \mathcal{A}_{jikl}(x) = \mathcal{A}_{ijlk}(x) = \mathcal{A}_{klij}(x), \quad \forall 1 \leq i, j, k, l \leq d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_{\mathcal{A}} > 0$ such that for a.e. $x \in \Omega$

$$s : \mathcal{A}(x) : s \geq \alpha_{\mathcal{A}} (s : s) \quad \forall s \in \mathbb{S}^d.$$

Moreover, there exists $C_{\mathcal{A}} > 0$ such that for a.e. $x \in \Omega$

$$|\mathcal{A}_{ijkl}(x)| \leq C_{\mathcal{A}} \quad \forall 1 \leq i, j, k, l \leq d.$$

The *internal* and *surface forces* in the body $\bar{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 2.8 (Cauchy Stress Tensor). *We define the (linear) Cauchy stress tensor by*

$$\sigma : H^s(\Omega; \mathbb{R}^d) \rightarrow H^{s-1}(\Omega; \mathbb{S}^d), \quad u \mapsto \mathcal{A} : \varepsilon(u),$$

where $\mathcal{A}$ is a sufficiently smooth elasticity tensor field, such that σ is well-defined.

In the case $s = 1$, it is sufficient to require $\mathcal{A}_{ijkl} \in L^\infty(\Omega)$ for every $1 \leq i, j, k, l \leq d$ for $\sigma(u) = \mathcal{A} : \varepsilon(u)$ to be part of $L^2(\Omega; \mathbb{S}^d)$ for every $u \in H^1(\Omega; \mathbb{R}^d)$. Later on, higher order Sobolev spaces are asked for, which requires higher regularity of $\mathcal{A}$. For example, for $3/2 < s \leq 2$ it is sufficient if $\mathcal{A}_{ijkl} \in W^{1,\infty}(\Omega)$ for every $1 \leq i, j, k, l \leq d$, while constant $\mathcal{A}$ always lead to well-defined stress tensors σ .

The function $f \in L^2(\Omega; \mathbb{R}^d)$ may describe a *volume force density* and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ may represent *surface loads* on the Neumann boundary Γ_N . With the basic principle that work is force times displacement, it is reasonable to define the linear functional

$$\ell(v) := \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad v \in H^1(\Omega; \mathbb{R}^d), \quad (2.1)$$

describing the *virtual work of external forces*. As the internal forces are described by the stress $\sigma(u) = \mathcal{A} : \varepsilon(u)$, the same principle suggests the symmetric bilinear form

$$a(u, v) := \int_{\Omega} \sigma(u) : \varepsilon(u) \, dx, \quad u, v \in H^1(\Omega; \mathbb{R}^d). \quad (2.2)$$

as a representative for the *virtual work of internal forces*. Equilibrium between both reads $a(u, v) = \ell(v)$ for all $v \in H^1(\Omega; \mathbb{R}^d)$, which already looks like a weak formulation but at this point, only the Neumann boundary conditions on Γ_N are imposed. In the following, we want to prove standard properties of a and ℓ that are needed later on. For this, we will need the following inequalities. The first inequality is a result of *Korn's second inequality* ([1, Theorem 3.3] and [1, Section 3.5.2]) and the *Peetre–Tartar Lemma* ([1, Lemma 3.5]). Therefore only the proof of the second one is given.

Lemma 2.1. *There exists a constant $C_K > 0$ such that*

$$C_K^{-1} \|v\|_{1,\Omega} \leq \|\varepsilon(v)\|_{0,\Omega} \leq \|v\|_{1,\Omega} \quad \forall v \in H^1(\Omega; \mathbb{R}^d).$$

Proof. We prove the second inequality. For all $v \in H^1(\Omega; \mathbb{R}^d)$ we have

$$|\varepsilon(v)(x)| = \left| \frac{1}{2} (\nabla v(x) + \nabla v^\top) \right| \leq \frac{1}{2} (|\nabla v(x)| + |\nabla v(x)^\top|) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|_{0,\Omega}^2 = \int_{\Omega} |\varepsilon(v)(x)|^2 \, dx \leq \int_{\Omega} |\nabla v(x)|^2 \, dx = \|\nabla v\|_{0,\Omega}^2 \leq \|v\|_{1,\Omega}^2,$$

where the last inequality holds due to

$$\|v\|_{1,\Omega}^2 = \|v\|_{0,\Omega}^2 + \|\nabla v\|_{0,\Omega}^2.$$

■

Proposition 2.3. *The symmetric bilinear form $a : H^1(\Omega; \mathbb{R}^d) \times H^1(\Omega; \mathbb{R}^d) \rightarrow \mathbb{R}$ defined in (2.2) is*

i) *bounded, i.e. for all $v, w \in H^1(\Omega; \mathbb{R}^d)$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \quad \text{with} \quad C_1 = C_{\mathcal{A}},$$

ii) *V-elliptic, i.e. for all $v \in H^1(\Omega; \mathbb{R}^d) \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \quad \text{with} \quad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined in (2.1) is bounded, i.e. for all $v \in H^1(\Omega; \mathbb{R}^d)$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \quad \text{with} \quad C_3 = \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}) \quad (2.3)$$

Proof. Let $v, w \in H^1(\Omega; \mathbb{R}^d)$ be arbitrary. Then

$$|a(v, w)| \leq \|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega}$$

by the Cauchy–Schwarz inequality. With Definition 2.7 there exists $C_{\mathcal{A}} > 0$ such that

$$\|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}.$$

Combining this with the second inequality from Lemma 2.1 we have

$$|a(v, w)| \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega} \leq C_{\mathcal{A}} \|v\|_{1,\Omega} \|w\|_{1,\Omega}.$$

The ellipticity of a follows by

$$a(v, v) = \int_{\Omega} \varepsilon(v) : \mathcal{A} : \varepsilon(v) \, dx \geq \alpha_{\mathcal{A}} \int_{\Omega} \varepsilon(v) : \varepsilon(v) = \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2,$$

where $\alpha_{\mathcal{A}} > 0$ exists due to Definition 2.7. The first inequality of Lemma 2.1 finally yields

$$a(v, v) \geq \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2 \geq \alpha_{\mathcal{A}} \left(C_K^{-1} \|v\|_{1,\Omega} \right)^2 = \frac{\alpha_{\mathcal{A}}}{C_K^2} \|v\|_{1,\Omega}^2.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{0,\Omega} \|v\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \|\gamma v\|_{0,\Gamma_N} \quad \forall v \in H^1(\Omega; \mathbb{R}^d).$$

By the Trace Theorem 2.1 there exists $C_{\text{tr}} > 0$ such that

$$\|\gamma v\|_{0,\Gamma_N} \leq \|\gamma v\|_{0,\Gamma} \leq \|\gamma v\|_{\frac{1}{2},\Gamma} \leq C_{\text{tr}} \|v\|_{1,\Omega} \quad \forall v \in H^1(\Omega; \mathbb{R}^d).$$

Finally,

$$|\ell(v)| \leq (\|f\|_{0,\Omega} + C_{\text{tr}} \|F\|_{0,\Gamma_N}) \|v\|_{1,\Omega} \leq \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}) \|v\|_{1,\Omega}$$

for every $v \in H^1(\Omega; \mathbb{R}^d)$, which proves (2.3). ■

2.4 Admissible Displacements

We start by introducing the underlying Hilbert space V in which homogeneous dirichlet boundary conditions on Γ_D are already incorporated by

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \mid (\gamma v)|_{\Gamma_D} = 0 \right\} \subset H^1(\Omega; \mathbb{R}^d).$$

For $v \in V$ we define the *normal component* of the boundary displacement on Γ by $v_n := (\gamma v) \cdot n$. Here, since Γ is Lipschitz, $\Gamma \in \mathcal{C}^{0,1}$, the outward normal n belongs to $L^\infty(\Omega; \mathbb{R}^d)$ which yields $v_n \in H^{1/2}(\Gamma)$. The rigid foundation is located on the exterior side of Γ_C in direction of n . Hence the body is not allowed to move across Γ_C into the obstacle. This yields that the solution must satisfy the *nonpenetration* condition

$$u_n \leq 0 \quad \text{a.e. on } \Gamma_C.$$

This motivates the nonempty, closed and convex cone of *admissible displacements*

$$K := \{v \in V \mid v_n \leq 0 \text{ a.e. on } \Gamma_C\} \subset V. \quad (2.4)$$

Proposition 2.4 (K is a closed convex cone). *The set K defined in 2.4 is a nonempty, closed and convex cone in V .*

Proof. K is nonempty, since $0 \in K$. Let $\lambda \in [0, \infty)$ and $v \in K \subset V$. Since V is a vector space, we have $\lambda v \in V$. Furthermore,

$$(\lambda v)_n = \gamma(\lambda v) \cdot n = (\lambda \gamma v) \cdot n = \lambda(\gamma v \cdot n) = \lambda v_n.$$

Since $v_n \leq 0$ a.e. on Γ_C and $\lambda \geq 0$, we also have $\lambda v_n \leq 0$ a.e. on Γ_C , i.e. $\lambda v \in K$. Thus K is a cone in V . Let $v, w \in K$ and $\theta \in [0, 1]$ and set $u := \theta v + (1 - \theta)w \in V$. Then we have

$$u_n = (\gamma u) \cdot n = (\theta \gamma v + (1 - \theta) \gamma w) \cdot n = \theta(\gamma v \cdot n) + (1 - \theta)(\gamma w \cdot n) = \theta v_n + (1 - \theta)w_n$$

and $v_n, w_n \leq 0$ a.e. on Γ_C and $\theta, (1 - \theta) \geq 0$. Thus,

$$\theta v_n + (1 - \theta)w_n \leq 0 \quad \text{a.e. on } \Gamma_C,$$

i.e. $u \in K$, which makes K convex. To prove that K is closed, let $(v_k)_{k \in \mathbb{N}} \subset K$ with $v_k \rightarrow v$ in V . Since the trace operator $\gamma : V \rightarrow H^{1/2}(\Gamma; \Omega)$ is continuous, the mapping

$$T : V \rightarrow L^2(\Gamma_C), \quad T(w) := ((\gamma w) \cdot \mathbf{n})|_{\Gamma_C} = w_n|_{\Gamma_C},$$

is continuous as well. Hence,

$$(v_k)_n = T(v_k) \rightarrow T(v) = v_n \quad \text{in } L^2(\Gamma_C).$$

Since $(v_k)_n \leq 0$ a.e. on Γ_C for all k , there exists a subsequence $((v_{k_j})_n)_{j \in \mathbb{N}}$ such that

$$(v_{k_j})_n(x) \rightarrow v_n(x) \quad \text{for a.e. } x \in \Gamma_C.$$

Passing to the limit in the inequality $(v_{k_j})_n(x) \leq 0$, we obtain

$$v_n(x) \leq 0 \quad \text{for a.e. } x \in \Gamma_C.$$

Thus $v \in K$, so K is closed. ■

3 Energy Minimization and Well-Posedness

As states earlier in Section 2.3, equilibrium between the virtual work of internal and external forces suggest to look for $u \in H^1(\Omega; \mathbb{R}^d)$ such that $a(u, v) = \ell(v)$ for all $v \in H^1(\Omega; \mathbb{R}^d)$. Here, only the Neumann conditions are imposed. Thus, a restriction on V , which imposes the homogeneous Dirichlet boundary conditions on Γ_D and further a restriction on the closed, convex cone K of displacements $v \in V$ that fullfil the nonpenetration condition on Γ_C , is reasonable. Naturally, this yields the quadratic functional

$$\mathcal{J}(v) := \frac{1}{2}a(v, v) - \ell(v) = \frac{1}{2} \int_{\Omega} \sigma(v) : \varepsilon(v) \, dx - \left[\int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds \right] \quad (3.1)$$

on V describing the *potential energy* of the elastic body to be minimized over $K \subset V$ to minimize the difference between $a(u, v)$ and $\ell(v)$ for all $v \in V$ under the nonpenetration constraint. This is already the first formulation of Signorini's problem as the minimization problem of $\mathcal{J}$ over K .

Problem 3.1 (Signorini's Problem – Energy Minimization).

$$\text{Find } u \in K \text{ such that: } \quad u \in \arg \min_{v \in K} \mathcal{J}(v).$$

Here we have decided to note u as an element of those v that minimize $\mathcal{J}$ over K . The reason for this is the lack of evidence so far of the problem's well-posedness, which includes existence and uniqueness of a minimizer and stability in the sense of a continuous dependence on the given data. Thus, we now what to prove well-posedness of the above Problem 3.1. Every formulation of Signorini's problem, which is proven to be equivalent to the above formulation is then also proven to be well-posed. We start by the uniqueness, which follows directly by the standard arguments of strict convexity of $\mathcal{J}$ and the convexity of K . The latter was already proven in Proposition 2.4.

Lemma 3.1 (Strict Convexity of $\mathcal{J}$). *The Functional $\mathcal{J}$ defined in (3.1) is strictly convex.*

Proof. For $u, v \in V$ and $t \in [0, 1]$, we set $w_t := (1 - t)u + tv$. Then

$$a(w_t, w_t) = (1 - t)a(u, u) + ta(v, v) - t(1 - t)a(u - v, u - v)$$

and

$$\ell(w_t) = (1 - t)\ell(u) + t\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1 - t)\mathcal{J}(u) + t\mathcal{J}(v) - \frac{1}{2}t(1 - t)a(u - v, u - v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V -ellipticity of a we have $a(u - v, u - v) > 0$ and $t(1 - t) > 0$. This implies

$$\mathcal{J}(w_t) < (1 - t)\mathcal{J}(u) + t\mathcal{J}(v).$$

■

Proposition 3.1 (Uniqueness of Minimizer of $\mathcal{J}$). *If there exists a minimizer of the functional $\mathcal{J}$ defined in (3.1), then it is unique.*

Proof. Assume that there exist two minimizers $u_1, u_2 \in K$ of $\mathcal{J}$ on K , i.e.

$$\mathcal{J}(u_1) = \min_{v \in K} \mathcal{J}(v) = \mathcal{J}(u_2). \quad (3.2)$$

If $u_1 = u_2$ there is nothing to show. Hence assume $u_1 \neq u_2$. Since K is convex, for every $t \in (0, 1)$ the convex combination

$$w_t := (1 - t)u_1 + tu_2$$

belongs to K . By Lemma 3.1, $\mathcal{J}$ is strictly convex and therefore we have

$$\mathcal{J}(w_t) < (1 - t)\mathcal{J}(u_1) + t\mathcal{J}(u_2) \quad \forall t \in (0, 1).$$

Using (3.2), this yields

$$\mathcal{J}(w_t) < (1 - t) \min_{v \in K} \mathcal{J}(v) + t \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \mathcal{J}(v).$$

This contradicts the definition of the minimum, since $w_t \in K$. Therefore the assumption $u_1 \neq u_2$ is false and we conclude $u_1 = u_2$. $\blacksquare$

It remains to show the existence of at least one minimizer. For this, we verify *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$. The proof requires the following lemma. Weak convergence is denoted by $\rightharpoonup$.

Lemma 3.2 (Weak lower semicontinuity of the norm). *Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \rightarrow \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X , then*

$$\|x\|_X \leq \liminf_{j \rightarrow \infty} \|x_j\|_X.$$

Proof. Let $J : X \rightarrow X^{**}$ be the canonical embedding defined by

$$(Jx)(\varphi) := \varphi(x) \quad \forall \varphi \in X'.$$

By the Hahn–Banach theorem, J is an isometry, i.e. $\|Jx\|_{X^{**}} = \|x\|_X$ for all $x \in X$. Therefore,

$$\|x\|_X = \|Jx\|_{X^{**}} = \sup_{\substack{\varphi \in X' \\ \|\varphi\|_{X'} \leq 1}} |(Jx)(\varphi)| = \sup_{\substack{\varphi \in X' \\ \|\varphi\|_{X'} \leq 1}} |\varphi(x)|.$$

If $x_j \rightharpoonup x$ in X , then $\varphi(x_j) \rightarrow \varphi(x)$ for all $\varphi \in X'$ and hence

$$\|x\|_X = \sup_{\substack{\varphi \in X' \\ \|\varphi\|_{X'} \leq 1}} |\varphi(x)| \leq \liminf_{j \rightarrow \infty} \sup_{\substack{\varphi \in X' \\ \|\varphi\|_{X'} \leq 1}} |\varphi(x_j)| = \liminf_{j \rightarrow \infty} \|x_j\|_X.$$

$\blacksquare$

Lemma 3.3 ($\mathcal{J}$ is coercive and w.l.s.c.). *The functional $\mathcal{J}$ defined in (3.1) is*

i) *coercive on V , i.e.*

$$\mathcal{J}(v) \rightarrow \infty \quad \text{as} \quad \|v\|_{1,\Omega} \rightarrow \infty,$$

ii) *weakly lower semicontinuous on V , i.e. if $v_j \rightharpoonup v$ in V , then*

$$\mathcal{J}(v) \leq \liminf_{j \rightarrow \infty} \mathcal{J}(v_j).$$

Proof. By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'} \|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \geq \frac{\alpha}{2} \|v\|_{1,\Omega}^2 - \|\ell\|_{V'} \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega} \rightarrow \infty} \infty.$$

For $s \in L^2(\Omega; \mathbb{S}^d)$ set

$$\|s\|_{\mathcal{A}}^2 := \int_{\Omega} s : \mathcal{A} : s \, dx.$$

By uniform ellipticity of $\mathcal{A}$, the map $\|\cdot\|_{\mathcal{A}}$ defines a norm on $L^2(\Omega; \mathbb{S}^d)$ equivalent to $\|\cdot\|_{0,\Omega}$. Moreover, $a(v, v) = \|\varepsilon(v)\|_{\mathcal{A}}^2$. The mapping $v \mapsto \varepsilon(v)$ is linear and bounded from V to $L^2(\Omega; \mathbb{S}^d)$. Hence, if $v_j \rightharpoonup v$ in V , then

$$\varepsilon(v_j) \rightharpoonup \varepsilon(v) \quad \text{in } L^2(\Omega; \mathbb{S}^d).$$

Since norms are weakly lower semicontinuous by Lemma 3.2, we obtain

$$a(v, v) = \|\varepsilon(v)\|_{\mathcal{A}}^2 \leq \liminf_{j \rightarrow \infty} \|\varepsilon(v_j)\|_{\mathcal{A}}^2 = \liminf_{j \rightarrow \infty} a(v_j, v_j).$$

Furthermore, $\ell \in V'$ is weakly continuous, hence $\ell(v_j) \rightarrow \ell(v)$. Therefore,

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \leq \frac{1}{2} \liminf_{j \rightarrow \infty} a(v_j, v_j) - \lim_{j \rightarrow \infty} \ell(v_j) = \liminf_{j \rightarrow \infty} \mathcal{J}(v_j).$$

■

Proposition 3.2 (Existence of Minimizer of $\mathcal{J}$). *The functional $\mathcal{J}$ defined in 3.1 has at least one minimizer $u \in K$.*

Proof. Set

$$M := \inf_{v \in K} \mathcal{J}(v) \in [-\infty, \infty).$$

Since $0 \in K$ we have $M \leq \mathcal{J}(0) = 0$, hence M is finite from above. Let $\{v_j\}_{j \in \mathbb{N}} \subset K$ be a minimizing sequence, i.e. $\mathcal{J}(v_j) \rightarrow M$ as $j \rightarrow \infty$. By Lemma 3.3 i), $\mathcal{J}$ is coercive and thus, the sequence $\{v_j\}_{j \in \mathbb{N}}$ is bounded in V . Since V is a Hilbert space, there exist a subsequence $\{v_{j_k}\}_{k \in \mathbb{N}}$ and some $u \in V$ such that

$$v_{j_k} \rightharpoonup u \quad \text{in } V.$$

As $K \subset V$ is closed and convex, it is weakly closed in V . Hence $u \in K$. By Lemma 3.3 ii), $\mathcal{J}$ is weakly lower semicontinuity and thus we obtain

$$\mathcal{J}(u) \leq \liminf_{k \rightarrow \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v),$$

so u is a minimizer of $\mathcal{J}$ on K . ■

We collect the previous results in the following corollary and further give a standard stability estimation for the norm of the unique solution.

Corollary 3.1 (Well-posedness of Problem 3.1). *The functional $\mathcal{J}$ defined in 3.1 admits a unique minimizer $u \in K$. Moreover, the minimizer satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2C_K^2 \max\{1, C_{\text{tr}}\}}{\alpha_{\mathcal{A}}} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}). \quad (3.3)$$

Consequently, Problem 3.1 has a unique stable solution.

Proof. By Proposition 3.2 we have existence of a solution $u \in K$ to Problem 3.1 and by Proposition 3.1 it is unique. It remains to prove the a-priori bound. Since u is the unique minimizer, it holds

$$\mathcal{J}(u) \leq \mathcal{J}(0) = 0 \quad \implies \quad \frac{1}{2} a(u, u) \leq \ell(u).$$

By Proposition 2.3 ii) we have $a(u, u) \geq \alpha_{\mathcal{A}}/C_K^2 \|u\|_{1,\Omega}^2$ and thus

$$\frac{\alpha_{\mathcal{A}}}{2C_K^2} \|u\|_{1,\Omega}^2 \leq \frac{1}{2} a(u, u) \leq |\ell(u)| \leq \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}) \|u\|_{1,\Omega},$$

where the last inequality is (2.3). Dividing both sides by $\|u\|_{1,\Omega}$ yields the desired estimate (3.3). For $u = 0$ the estimate is trivial. ■

4 Weak Formulations

Finally, we turn to the *weak formulation* of Signorini's problem. Unlike the weak formulation of conventional simple boundary value problems, here we do not end up with a variational equation, but with a variational inequality. This is derived by a first-order optimality condition on the functional $\mathcal{J}$ under the constraints of K , i.e. a condition on the *Fréchet derivative* of $\mathcal{J}$. Therefore to fix notations, we start with a definition.

Definition 4.1 (Fréchet derivative). *Let X and Y be normed vector spaces and let*

$$f : U \subseteq X \rightarrow Y$$

*be a function defined on an open subset U of X . Then f is called **Fréchet differentiable** at $x \in U$ if there exists a bounded linear operator $A(x) : X \rightarrow Y$ such that*

$$\lim_{\|h\|_X \rightarrow 0} \frac{\|f(x+h) - f(x) - A(x)[h]\|_Y}{\|h\|_X} = 0. \quad (4.1)$$

*If $A(x)$ exists, it is unique and is called the **Fréchet derivative** of f at x . We write*

$$f'(x)[h] = A(x)[h] \quad \forall h \in X.$$

Finally, we prove Fréchet differentiability of the energy functional

$$\mathcal{J}(v) := \frac{1}{2}a(v, v) - \ell(v), \quad v \in V$$

and provide an explicit formula for its Fréchet derivative.

Lemma 4.1 (Fréchet differentiability of $\mathcal{J}$). *The functional $\mathcal{J}$ defined in (3.1) is Fréchet differentiable on V . Its derivative at $u \in V$ is given by*

$$\mathcal{J}'(u)[h] = a(u, h) - \ell(h) \quad \forall h \in V. \quad (4.2)$$

Proof. Let $u, k \in V$. Using bilinearity of a and linearity of ℓ , we obtain

$$\begin{aligned} \mathcal{J}(u+k) - \mathcal{J}(u) &= \frac{1}{2}(a(u+k, u+k) - a(u, u)) - \ell(k) \\ &= \frac{1}{2}(a(u, u) + 2a(u, k) + a(k, k) - a(u, u)) - \ell(k) \\ &= a(u, k) - \ell(k) + \frac{1}{2}a(k, k). \end{aligned}$$

Hence

$$\mathcal{J}(u+k) - \mathcal{J}(u) - (a(u, k) - \ell(k)) = \frac{1}{2}a(k, k).$$

Define $A(u) : V \rightarrow \mathbb{R}$ by

$$A(u)[h] := a(u, h) - \ell(h).$$

Since a and ℓ are bounded, $A(u)$ is linear and bounded. Moreover,

$$\frac{|\mathcal{J}(u+k) - \mathcal{J}(u) - A(u)[k]|}{\|k\|_{1,\Omega}} = \frac{1}{2} \frac{|a(k, k)|}{\|k\|_{1,\Omega}} \leq \frac{1}{2} C_{\mathcal{A}} \|k\|_{1,\Omega} \xrightarrow{\|k\|_{1,\Omega} \rightarrow 0} 0,$$

by boundedness of a , cf. Proposition 2.3. Therefore, $\mathcal{J}$ is Fréchet differentiable at u with

$$\mathcal{J}'(u)[h] = A(u)[h] = a(u, h) - \ell(h) \quad \forall h \in V.$$

■

4.1 Variational Inequality as an Optimality Condition

Proposition 4.1 (First-order optimality condition on convex sets). *A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v - u] \geq 0 \quad \forall v \in K. \quad (4.3)$$

Proof. Assume first that $u \in K$ minimizes $\mathcal{J}$ over K and let $v \in K$. By convexity of K we have $u + t(v - u) \in K$ for all $t \in [0, 1]$. Define $\phi(t) := \mathcal{J}(u + t(v - u))$. Then ϕ attains its minimum at $t = 0$, hence $\phi'(0^+) \geq 0$, where $\phi'(0^+)$ denotes the right-sided derivative at $t = 0$. Using the chain rule and Fréchet differentiability of $\mathcal{J}$ we obtain

$$\phi'(0^+) = \mathcal{J}'(u)[v - u] \geq 0.$$

Conversely, assume (4.3). Let $v \in K$ be arbitrary and set $h := v - u$. By Taylor expansion of the quadratic functional $\mathcal{J}$ we have

$$\mathcal{J}(u + h) - \mathcal{J}(u) = \mathcal{J}'(u)[h] + \frac{1}{2}a(h, h).$$

By (4.3), $\mathcal{J}'(u)[h] \geq 0$, and since $a(h, h) \geq 0$ we conclude $\mathcal{J}(v) \geq \mathcal{J}(u)$ for all $v \in K$. Hence u is a minimizer over K . ■

Combining Lemma 4.1 and Proposition 4.1, $u \in K$ solves Problem 3.1 if and only if

$$\mathcal{J}'(u)[v - u] = a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K.$$

This is the first weak formulation of Signorini's problem as a variational inequality.

Problem 4.1 (Signorini's Problem – Weak Formulation 1).

$$\text{Find } u \in K \text{ such that:} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

Well-posedness of Problem 4.1 is already ensured, since it is equivalent to the minimization Problem 3.1 by Proposition 4.1 in the sense that each solution to the one of the problems is a solution to the other problem. The solution is unique due to Corollary 3.1 with the stability estimate (3.3). Another useful equivalent formulation is the following.

Problem 4.2 (Signorini's Problem – Weak Formulation 2).

$$\text{Find } u \in K \text{ such that:} \quad \begin{cases} a(u, u) = \ell(u), \\ a(u, v) \geq \ell(v) \end{cases} \quad \forall v \in K.$$

Proposition 4.2 (Equivalence of Problem 4.1 and Problem 4.2). *Any solution $u \in K$ to Problem 4.1 is also a solution to Problem 4.2 and vice versa.*

Proof. Let $u \in K$ solve Problem 4.2. Then

$$a(u, v - u) = a(u, v) - a(u, u) = a(u, v) - \ell(u) \geq \ell(v) - \ell(u) = \ell(v - u).$$

Assuming that $u \in K$ solves Problem 4.1 and testing $v = 0$ yields

$$a(u, -u) \geq \ell(-u) \quad \Longleftrightarrow \quad -a(u, u) \geq -\ell(u),$$

which directly implies

$$a(u, u) \leq \ell(u).$$

However, testing with $v = 2u$ yields $a(u, u) \geq \ell(u)$ and thus $a(u, u) = \ell(u)$. Furthermore, Problem 4.1 reads

$$a(u, v - u) = a(u, v) - a(u, u) \geq \ell(v - u) = \ell(v) - \ell(u),$$

which directly implies

$$a(u, v) \geq a(u, u) + \ell(v) - \ell(u).$$

Plugging in $a(u, u) = \ell(u)$ yields $a(u, v) \geq \ell(v)$. ■

4.2 Lipschitz Dependence of the Displacement on the Forces

We close this section by providing another stability result on the dependency of solutions to varying source terms as input data. This result is very similar to the stability estimate provided in Corollary 3.1. Its proof however, is way more comfortable at this point, where the weak formulations are available.

Proposition 4.3 (Lipschitz dependence on the data). *For $i \in \{1, 2\}$, let $f_i \in L^2(\Omega; \mathbb{R}^d)$ and $F_i \in L^2(\Gamma_N; \mathbb{R}^d)$, and let $\ell_i \in V'$ be the corresponding load functional, i.e.*

$$\ell_i(v) := \int_{\Omega} f_i \cdot v \, dx + \int_{\Gamma_N} F_i \cdot \gamma v \, ds, \quad v \in V.$$

Moreover, let $u_i \in K$ be the unique solution of Problem 4.1 associated with ℓ_i , i.e.

$$a(u_i, v - u_i) \geq \ell_i(v - u_i) \quad \forall v \in K.$$

Then the solution depends Lipschitz continuously on the load functional, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2}{\alpha_{\mathcal{A}}} \|\ell_1 - \ell_2\|_{V'}. \quad (4.4)$$

In particular, it depends Lipschitz continuously on the data (f_i, F_i) in the sense that

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2 \max\{1, C_{\text{tr}}\}}{\alpha_{\mathcal{A}}} (\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}). \quad (4.5)$$

Proof. For u_1 , choose $v = u_2 \in K$ in Problem 4.1. Then $a(u_1, u_2 - u_1) \geq \ell_1(u_2 - u_1)$. Similarly, choosing $v = u_1 \in K$ for u_2 yields $a(u_2, u_1 - u_2) \geq \ell_2(u_1 - u_2)$. Adding both inequalities and using symmetry of a , we obtain

$$a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2).$$

By Proposition 2.3, the bilinear form a is V -elliptic with constant $\alpha_{\mathcal{A}}/C_K^2$. Hence

$$\frac{\alpha_{\mathcal{A}}}{C_K^2} \|u_1 - u_2\|_{1,\Omega}^2 \leq a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2) \leq \|\ell_1 - \ell_2\|_{V'} \|u_1 - u_2\|_{1,\Omega}.$$

If $u_1 = u_2$, there is nothing to prove. Otherwise, dividing by $\|u_1 - u_2\|_{1,\Omega}$ yields (4.4). Moreover,

$$(\ell_1 - \ell_2)(v) = \int_{\Omega} (f_1 - f_2) \cdot v \, dx + \int_{\Gamma_N} (F_1 - F_2) \cdot \gamma v \, ds \quad \forall v \in V.$$

Therefore, Proposition 2.3 applied to the functional $\ell_1 - \ell_2$ implies

$$\|\ell_1 - \ell_2\|_{V'} \leq \max\{1, C_{\text{tr}}\} (\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}).$$

Combining this with (4.4), we obtain (4.5). ■

5 Strong Formulations

5.1 Stress Regularity and Normal Traces in $H(\text{div}; \Omega)$

5.2 A Hybrid Formulation

5.3 Normal/Tangential Splitting and Contact Trace Spaces

5.4 Formulation with KKT Contact Conditions

5.5 Classical Strong Formulation under Additional Regularity

6 Mixed Formulation

7 Outlook

A References

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